

PAPER_742: Sombrero Galaxy MUGE — Dust Lane Drag Term D_dust

Author: Daniel T. Murphy

Framework: Universal Quantum Field Superconductive Framework (UQFF)

Session: 180 continuation | v5.38

Date: 2025

CP4 Class: #326 — SombreroGalaxyDustMUGECalculator

Abstract

The Sombrero Galaxy (M104) presents a unique astrophysical challenge: its prominent equatorial dust lane exerts measurable drag on bulge dynamics and star formation near the central black hole ($\sim 10^6 M_\odot$). This paper derives the MUGE for the Sombrero Galaxy incorporating the new D_{dust} environmental term, extending the standard UQFF formalism to include optically thick dust lane physics via the F_{env} operator.

1. Introduction

M104 (NGC 4594), the Sombrero Galaxy, is a lenticular/Sa galaxy at 28 Mpc with one of the most prominent dust lanes in the local universe. Its structure features a massive bulge, a flat disk embedded in dust, and a $10^6 M_\odot$ SMBH. The dust lane creates a unique gravitational environment where extinction, drag forces, and angular momentum exchange compete with bulge and halo dynamics.

Hubble parameters for M104: - $M_{\text{visible}} \sim 8 \times 10^{11} M_\odot$ - $M_{\text{BH}} \sim 10^6 M_\odot$ - Distance ~ 28 Mpc - $r_{\text{galaxy}} \sim 50$ kpc - $\rho_{\text{dust}} \sim 10^{-2} \text{ kg/m}^3$ (dust lane) - $B \sim 5 \times 10^{-4} \text{ T}$ (typical galactic field)

2. Sombrero Galaxy MUGE

$$\begin{aligned} g_{\text{Sombrero}}(r,t) = & (G \cdot M)/r^2 \cdot (1+H(z) \cdot t) \cdot (1-B/B_{\text{crit}}) \\ & + (G \cdot M_{\text{BH}})/r_{\text{BH}}^2 \\ & + (U_{\text{g1}} + U_{\text{g2}} + U_{\text{g3}} + U_{\text{g4}}) \\ & + U_{\text{i}} \\ & + (\Lambda \cdot c^2/3) \\ & + (\hbar/\sqrt{\Delta x \cdot \Delta p}) \cdot (\dot{H} \cdot dV) \cdot (2/t_{\text{Hubble}}) \\ & + \rho_{\text{fluid}} \cdot V \cdot g \\ & + (M_{\text{vis}} + M_{\text{DM}}) \cdot (\ddot{\phi} + 3 \cdot G \cdot M/r^3) \\ & + D_{\text{dust}} \end{aligned}$$

3. D_dust — Dust Lane Drag Term

The novel term D_dust models the retarding force exerted by the optically thick equatorial dust lane on bulge stellar orbits and gas dynamics:

$$D_{\text{dust}} = -k_{\text{dust}} \cdot \rho_{\text{dust}} \cdot v_{\text{orbit}}^2 \cdot A_{\text{cross}} / r$$

k_{dust} = dust-gas coupling coefficient (~ 0.5)
 ρ_{dust} = dust lane mass density ($\sim 10^{-2}$ kg/m³ for M104)
 v_{orbit} = local orbital velocity (m/s)
 A_{cross} = cross-sectional area of interaction (m²)
 r = radial distance from center (m)

For the Sombrero bulge region ($r \sim 1\text{--}5$ kpc):

$$D_{\text{dust}} \sim 5 \times 10^{-12} \text{ m/s}^2 \quad (\text{retarding contribution})$$

This is comparable in magnitude to the dark matter perturbation term, making it non-negligible for bulge kinematics near $r < 3$ kpc.

4. Black Hole Contribution

The Sombrero's SMBH ($M_{\text{BH}} \sim 10^6 M_{\odot}$) dominates within $r < 1$ kpc:

$$g_{\text{BH}} = G \cdot M_{\text{BH}} / r_{\text{BH}}^2$$
$$g_{\text{BH}}(1 \text{ kpc}) \sim 2.4 \times 10^{-12} \text{ m/s}^2$$

5. U_g Terms (Sombrero Configuration)

$$U_{\text{g1}} = \rho_{\text{dipole}} \cdot B \quad (\text{AGN magnetic dipole})$$
$$\rho_{\text{dipole}} = I \cdot A \cdot \omega_{\text{spin}} \sim 10^{-1} \text{ A} \cdot \text{m}^2$$

$$U_{\text{g2}} = B_{\text{super}}^2 / (2 \cdot \epsilon_0) \quad (\text{aether superconductor field})$$
$$B_{\text{super}} = \epsilon_0 \cdot H_{\text{aether}}, H_{\text{aether}} \sim 10^4 \text{ A/m}$$

$$U_{\text{g3}} = G \cdot M_{\text{ext}} / r_{\text{ext}}^2 \quad (\text{companion galaxy influence})$$

$$U_{\text{g4}} = k_4 \cdot \rho_{\text{vac}} [\text{SCm}] \cdot (M_{\text{BH}}/d_g) \cdot e^{-(t/\tau)} \cdot \cos(\omega t) \cdot (1+f_{\text{feedback}})$$
$$k_4 = 1.0$$

6. Environmental Forcing for M104

$$F_{\text{env}}(t) = F_{\text{dust}} + F_{\text{BH}} + F_{\text{cosmo}}$$

$$\begin{aligned} F_{\text{dust}} &= D_{\text{dust}} / g_{\text{Newtonian}} && \text{(dust drag fraction)} \\ F_{\text{BH}} &= G \cdot M_{\text{BH}} / (r^2 \cdot g_{\text{Newtonian}}) && \text{(BH contribution fraction)} \\ F_{\text{cosmo}} &= H(z) \cdot t && \text{(cosmological expansion)} \end{aligned}$$

7. Full Equation Solutions

For standard bulge parameters ($r = 5$ kpc, $t = 0$): $-g_{\text{Newton}} \sim 4.5 \times 10^{-1} \text{ m/s}^2$
- $H(z) \cdot t$ correction $+0.35\%$ - $(1-B/B_{\text{crit}}) \sim 1$ (negligible for $B \ll B_{\text{crit}}$) -
 $D_{\text{dust}} \sim -5 \times 10^{-12} \text{ m/s}^2$ (-1.1% correction) - $g_{\text{Sombrero}} \sim 4.5 \times 10^{-1} - 5 \times 10^{-12}$
+ quantum + DM terms

8. Astrophysical Significance

The D_{dust} term explains: 1. **Reduced star formation efficiency** in the dust lane compared to dust-free bulges 2. **Kinematic misalignment** of dust-embedded stars vs. outer halo stars 3. **SMBH growth rate suppression** through dust-mediated angular momentum transfer 4. **Mid-infrared brightness** excess near the dust lane (consistent with Spitzer observations)

9. Conclusion

The Sombrero Galaxy MUGE extends UQFF to dust-dominated lenticular/Sa galactic environments via the D_{dust} term. The dust lane drag force amounts to $\sim 1\%$ correction on bulge dynamics but significantly affects the SMBH growth environment and inner kinematics. Future integration with Spitzer/JWST dust mass measurements can refine D_{dust} for M104 and analogous systems.

Copyright - Daniel T. Murphy, daniel.murphy00@gmail.com. UQFF Framework. PAPER_742, CP4 class #326. Session 180 continuation v5.38.